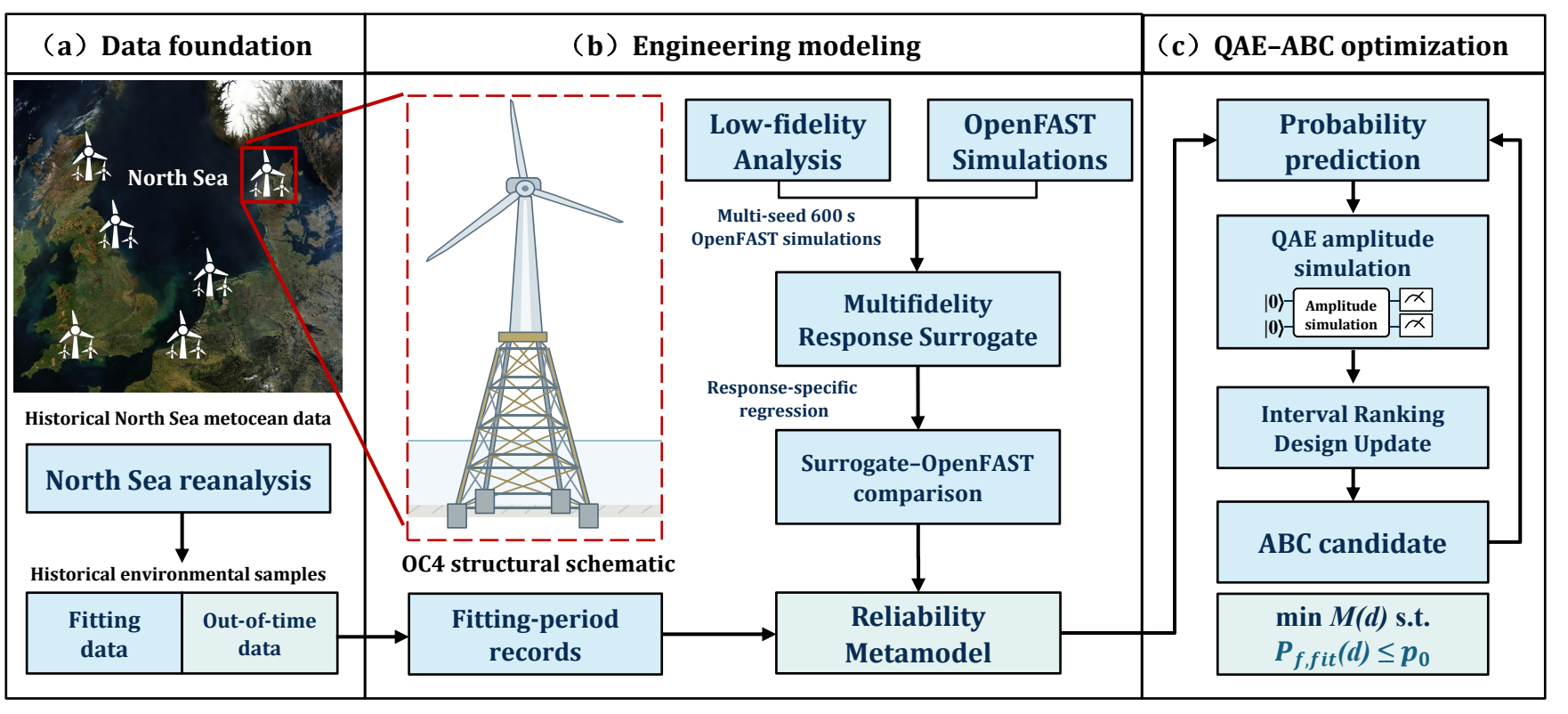

(a) Data foundation
(b) Engineering modeling
(c) QAE–ABC optimization
North Sea
Historical North Sea metocean data
North Sea reanalysis
Historical environmental samples
Fitting data
Out-of-time data
OC4 structural schematic
Fitting-period records
Low-fidelity Analysis
OpenFAST Simulations
Multi-seed 600 s OpenFAST simulations
Multifidelity Response Surrogate
Response-specific regression
Surrogate–OpenFAST comparison
Reliability Metamodel
Probability prediction
QAE amplitude simulation
|0⟩
|0⟩
Amplitude simulation
Interval Ranking Design Update
ABC candidate
min M(d) s.t.
$P_{f,fit}(d) \le p_0$